

Introduction:

Though biologically reproductive health means the healthy and normal functioning of reproductive organs, the world health organization has defined reproductive health defined as **the total well-being in all aspects of reproduction, i.e., physical, emotional, behavioural and social**. In the present chapter the problems in reproductive health and strategies to overcome the same are discussed.

Reproductive Health – Problems and Strategies

- India was among the first countries in the world to ensure reproductive health to all, and since 1951, successive governments have introduced a number of programmes to ensure this, e.g., **family planning** and **RCH** (Reproductive and Child Healthcare) programmes.

The major aims and objective of RCH programme (General strategies to tackle reproductive problems and achieve reproductive health)

- To create awareness among people about the various aspects of reproductive health.
- Several audio-visual aids and pamphlets have been developed by both governmental and non-governmental organisations.
- Awareness is being created among school children by introducing sex education in schools.
- Educating people, especially fertile couples and those in marriageable age group, about available birth control options, care of pregnant mothers, post-natal care of the mother and child, importance of breast feeding, equal opportunities for the male and the female child.
- Awareness of problems due to uncontrolled population growth, social evils like sex – abuse and sex related crimes
- It is essential to provide medical assistance to people with problems related to STDs (Sexually Transmitted Diseases), pregnancy, contraception and infertility, especially in villages.
- Statutory ban on **Amniocentesis** (foetal sex determination test based on the chromosomal pattern in the amniotic fluids surrounding the developing embryo) to check female foeticide.

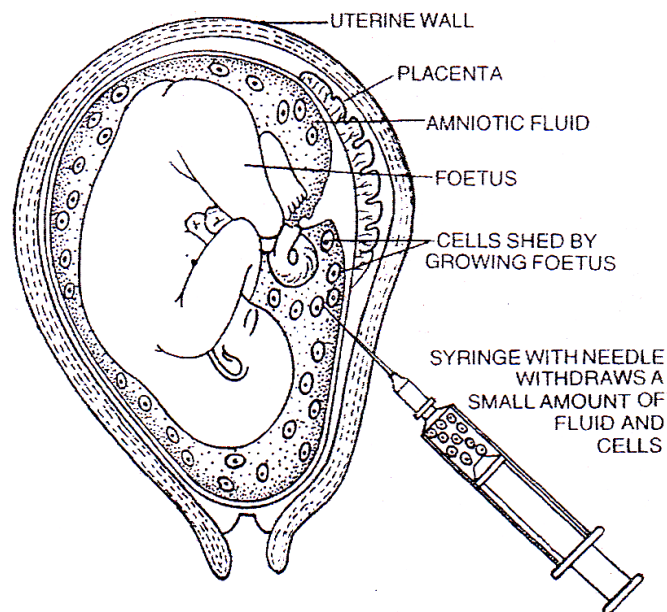


Figure : Amniocentesis

- Research on various reproduction-related areas are encouraged and supported by governmental and non-governmental agencies to find out new methods and/or to improve upon the existing ones (*'Saheli'*—a new oral contraceptive for the females—was developed by scientists at Central Drug Research Institute (CDRI) in Lucknow, India)

An improved reproductive health of the society indicates

- Better awareness about sex related matters
- Increased number of medically assisted deliveries and better post-natal care leading to decreased maternal and infant mortality rates.
- Increased number of couples with small families
- Better detection and cure of STDs and overall increased medical facilities for all sex related problems.

POPULATION EXPLOSION

- Improvement in the quality of life due to all-round development (better health facilities and improved conditions of living) has led to a large increase in the world population.
- The world population which was around 2 billion (2000 million) in 1900 rocketed to about 6 billion by 2000.
- Indian population was approximately 350 million at the time of independence reached to the billion and crossed by 1 billion by May 2000. That means, every sixth person in the world is an Indian.

The causes of population explosion are:

- Decline in death rate, including **maternal mortality rate** (MMR) and **infant mortality rate** (IMR)
- Increase in the birth rate, better health facilities and improved condition of living.
- Increase in the number of people in the reproductive age
- Population explosion causes a scarcity of every basic need.
- Therefore, it has become essential to check this increase in population.

Steps to overcome population explosion

- To motivate smaller families by using various contraceptive methods. You might have seen advertisement in the media as well as poster/bill, etc., showing a happy couple with two children with a slogan "**HUM DO HAMARE DO**" (We two, our two)
- One-child norm.
- Increasing the marriageable age of the female to 18 and male to 21 years.
- Incentives given to couples with small families.
- Promoting contraceptive methods are some of the ways to check population explosion.

BIRTH CONTROL

- It is essential to ensure birth control to check population explosion.
- Successive governments have come out with various programmes that encourage smaller families by means of various contraceptive methods.
- Contraceptive methods helps in preventing the unwanted pregnancies.
- **An ideal contraceptive should be –**
 - User friendly.
 - Easily available.
 - Effective and reversible with no side effect.
 - It should no way interfere with the sexual desire and/or the sexual act of the user.

Types of contraceptive methods

Contraceptive methods are of two types:

- 1. Natural means of contraception**
- 2. Artificial means of contraception**

1. Natural Methods

- It work on the principle of avoiding chances ovum and sperms meeting.
- Natural methods includes.
 - i) Periodic Abstinence or Rhythm Method** (temporary avoidance of sex): In this method the couples avoid or abstain from coitus from day 10 to 17th day of the menstrual cycle when ovulation could be expected. As chances of fertilization are very high during this period, it is called the **fertile period**.
 - ii) Coitus Interruptus or Withdrawal**: Male partner withdraw his penis from the vagina just before ejaculation.
 - iii) Lactational Amenorrhea Method (LAM)**: It is the absence of menstruation during lactation. So, in the lactation period, the chances of conception are almost zero. This method has been reported to be **effective only upto a maximum period of six months** following parturition.

Importance of Natural contraceptive method.

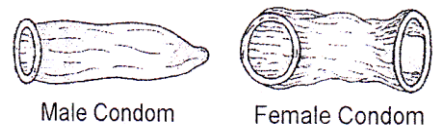
- As no medicines or devices are used
- No side effects.

2. Artificial Methods

- Physically preventing the coming together of ovum and sperm
- Artificial methods includes – Barriers, Intra uterine devices (IUDs), Oral contraceptive pills, surgical methods.
- i. Barriers**: In this method sperm and ovum are prevented from physically meeting with the help of barrier. Such methods are available for both males and females. Examples :- Condoms, Diaphragms, cervical caps and vaults

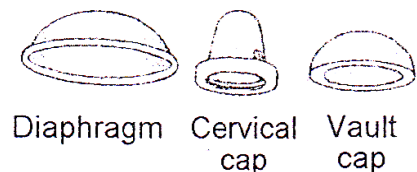
Condoms

- Condoms are barrier made of thin rubber / latex sheath.
- Condoms are used to cover penis in the male or vagina and cervix in the female, just before the coitus so that the ejaculated semen would not enter into the female reproductive tract. This can prevent the conception.
- **"Nirodh"** is a popular brand of condom for the male.
- Use of condom has increased in recent years due to its additional benefit of protecting the user from contracting STDs and AIDS.
- Both the male and the female condoms are disposable and be self-inserted and thereby gives privacy to the user.



Diaphragms, Cervical caps and Vaults.

- These are also barriers made of rubber that are inserted into the female reproductive tract to cover the cervix during coitus.
- They prevent conception by blocking the entry of sperms through the cervix.
- They are reusable.
- Spermicidal creams, jellies and foams are usually used along with these barriers to increase their contraceptive efficiency.



ii. Intra uterine devices (IUDs):

- These devices are inserted by doctors or expert nurses in the uterus through vagina.
- IUDs increase phagocytosis of sperm within the uterus.

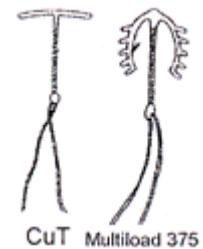
- IUDs are ideal contraceptives for the female who want to delay pregnancy and / or space children.
- It is one of most widely accepted methods of contraception in India
- These Intra Uterine devices are presently available as the
 - a) Non-medicated IUDs
 - b) Copper releasing IUDs
 - c) Hormone releasing IUDs

a) Non – Medicated IUDs

- Non-medicated IUD cause a sterile inflammatory response by producing a tissue injury of minor degree but sufficient enough to be spermicidal.
- Ex – Lippes loop

b) Copper releasing IUDs

- They release Cu ions to suppress the sperm motility and the fertilising capacity of sperms.
- Examples :- CuT, Cu7, Multiload 375

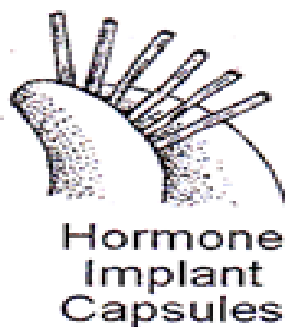


c) Hormone releasing IUDs

- They make uterus unsuitable for implantation and cervix hostile to the sperm.
- Examples :- Progestasert, LNG-20 (Levonogestrel 20 µg/day) (Trade name – Minera)

iii. Oral contraceptive pills:

- These are used in the form of tablets and hence are popularly known as **Pills**
- Pills are very effective with lesser side effects and are well accepted by the females
- They contain hormones—either progesterone or progesterone–oestrogen combination.
- They inhibit ovulation and implantation as well as alter the quality of cervical mucus to prevent / retard entry of sperm.
- Pills have to be taken daily for a period of 21 days starting preferably within the first five days of menstrual cycle. After a gap of 7 days (during which menstruation occurs) or as to be repeated in the same pattern till the female desires to prevent conception.
- Progesterone alone or in combination with estrogen can also be used by females as injections (Depo Provera) or implants (Implanon, Levenogestrel norplant) under the skin. Their mode of action is similar to that of pills and their effective periods are much longer.

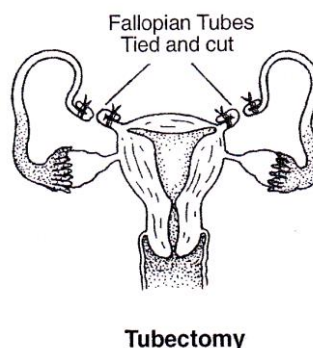
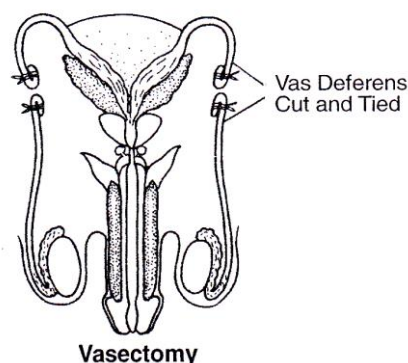


- Administration of progesterone or progesterone –estrogen combination or IUDs within 72 hours of coitus have been found to be very effective as emergency contraceptives as they could be used to avoid possible pregnancy due to rape or causal unprotected intercourse.
- Examples- **Saheli** – the new oral contraceptive for the females contains a non-steroidal preparation. It is a 'once a week' pill with very few side effects and high contraceptive value.

iv. Surgical methods:

- Also called **sterilization**.
- This method is generally advice for the males and female as a terminal method to prevent any more pregnancies.

- **Vasectomy** – It is the sterilization procedure in males. It involves the removal of small part of the vas deferens or tied up through a small incision on the scrotum.
- **Tubectomy** – It is the sterilization procedure in females. It involves the removal of the fallopian tube or tied up through a small incision in the abdomen or through vagina.



NOTE: - It needs to be emphasized that the selection of a suitable contraceptive method and its use should always be undertaken in consultation with qualified medical professionals. One must also remember that contraceptives are not regular requirements for the maintenance of reproductive events, i.e., conception / pregnancy. However, their possible ill effects like nausea, abdominal pain, breakthrough bleeding, irregular menstrual bleeding or even breast cancer, though not very significant, should not be totally ignored.

Difference between vasectomy and tubectomy

	Vasectomy		Tubectomy
1.	It is surgical sterilisation technique for the males.	1.	It is surgical sterilisation technique for the females.
2.	The two vasa deferentia are interrupted by giving cuts.	2.	The two oviducts are interrupted by either ligation or cuts.
3.	Transport of sperms is blocked.	3.	Transport of ova is blocked.

MEDICAL TERMINATION OF PREGNANCY (MTP) / INDUCED ABORTION

- Intentional or voluntary termination of pregnancy before full term is called medical termination of pregnancy (MTP) or induced abortion.
- Nearly 45 to 50 million MTPs are performed in a year all over the world which accounts to 1/5th of the total number of conceived pregnancies in a year.
- Government of India legalized MTP in 1971 with some strict conditions to avoid its misuse. Such restriction are all the more important to check indiscriminate and illegal female foeticides which are reported to be high in India.

Healthy trends of MTP

- MTP is necessary to get rid of unwanted pregnancies either due to casual unprotected intercourse or failure of the contraceptive used during coitus or rapes.
- It is also essential in certain cases where continuation of the pregnancy could be harmful or even fatal either to the mother or to the foetus or both.

NOTE: - MTPs are considered relatively safe during the first trimester, i.e., upto 12 weeks of pregnancy. Second trimester abortions are much more riskier

Unhealthy trends of MTP

- Majority of MTPs are performed illegally by unqualified quacks which are not only unsafe but could be fatal to both for young mother and the foetus.
- Misuse of amniocentesis to determine the sex of the unborn child.
- If the foetus is found to be female, it is followed by MTP- this is totally against what is legal.

NOTE: - Effective counselling on the need to avoid unprotected coitus and the risk factors involved in illegal abortions as well as providing more health care facilities could reverse the mentioned unhealthy trend.

Sexually Transmitted Diseases (STDs) or Venereal Diseases (VD) or Reproductive Tract Infections (RTI)

- Diseases of infections which are transmitted through sexual intercourse are collectively called sexually transmitted diseases (STD) or venereal diseases (VD) or reproductive tract infections (RTI).
- Examples: - Gonorrhoea, syphilis, genital herpes, chlamydiasis, genital warts, trichomoniasis, hepatitis-B and HIV.

Mode of transmission

- Unsafe coitus with multiple partners or with infected persons.
- Infections like hepatitis-B and HIV can also be transmitted by sharing of injections like needles, surgical instruments, etc., with infected persons, transfusion of blood, or from an infected mother to the foetus too.

NOTE – except for hepatitis-B, genital herpes and HIV infections other diseases are completely curable if detected early and treated properly.

Symptoms

- **Early symptoms includes** - minor and include itching, fluid discharge, slight pain, swellings, etc., in the genital region. Infected females may often be asymptomatic and hence, may remain undetected for long. Absence or less significant symptoms in the early stages of infection and the social stigma attached to the STDs, deter the infected persons from going for timely detection and proper treatment.
- **Late symptoms includes** – abortions, still births, ectopic pregnancies (if the implantation of blastocyst occur other than the uterus is called ectopic pregnancies), **pelvic inflammatory diseases (PID)**, infertility or even cancer of the reproductive tract. Infection and the social stigma attached to the STDs, deter the infected persons from going for timely detection and proper treatment.

NOTE: - though all persons are vulnerable to these infections, their incidences are reported to be very high among persons in the age group of 15 – 24 years.

Prevention

- Avoid sex with unknown partners or multiple partners.
- Always use condoms during coitus.
- In case of doubt, go to a qualified doctor for early detection and get complete treatment if diagnosed with diseases.

INFERTILITY

- Inability of couples to produce children is called **infertility**.
- Many couples all over the world are unable to produce children.
- Some of the reasons for infertility are:
 - Congenital diseases
 - Drugs
 - Immunological and Psychological factors
- Specialised fertility clinics can help diagnose and treat infertility.
- The couples can be assisted to have children through techniques called **assisted reproductive technologies (ART)**.

Extra information on infertility

Causes of infertility in men

- Physical abnormalities of the penis
- **Impotency** – it is the inability of the male to attain or hold an erection of the penis long enough for normal intercourse.
- Endocrine disorders like hyposecretion of gonadotropic hormones
- Nonfunctional testes. It may be congenital or due to damage.
- **Cryptorchidism** (from Greek kryptos "hidden" and orchis "testis") - is the absence of one or both testes from the scrotum.
- Defective epididymis and vasdeference
- **Oligospermia** – deficiency in sperm count (< 20 million/ml) (normal value of sperm count is 50-150 million per ml of semen)
- **Azoospermia** – complete absence of sperm in the ejaculate
- **Asthenospermia** or **Asthenozoospermia** – the condition in which male has poor sperm motility

Causes of infertility in women

- Endocrine disorders or hormonal imbalance
- Injuries or disease of the ovaries
- Non-ovulatory menstrual cycle
- Blockage in the fallopian tube
- Non-canalization of vagina
- Improper development of uterus
- Tumor in the uterus

Assisted Reproductive Technologies (ART)

- **In vitro fertilisation (IVF):** Fertilisation takes place outside the body which is followed by **embryo transfer (ET)**.
- This method also popularly known as **Test Tube Baby**.
- The following techniques are included in IVF.

ZIFT (zygote intra fallopian transfer) - In ZIFT, the sperm from a donor male and the ova from a donor female are fused in the laboratory. The zygote so formed is transferred into the fallopian tube at the 8 blastomeres stage.

IUT (intra uterine transfer) - In this technique, embryos with more than 8 blastomeres are transferred into the uterus.

GIFT (gamete intra fallopian transfer) - In GIFT, ovum is collected from a donor and transferred to another female who cannot produce one, but can provide suitable environment for fertilization and further development.

ICSI (intra cytoplasmic sperm injection) - In this method, sperm is directly injected into the cytoplasm of the ovum.

AI (Artificial insemination) – In this technique, the semen collected from the husband or a donor is injected into the vagina or uterus. This cures the infertility arising from the inability of the male partner to ejaculate, or due to low sperm count.

SUMMARY

Reproductive health refers to a total well-being in all aspects of reproduction, i.e., physical, emotional, behavioural and social. Our nation was the first nation in the world to initiate various action plans at national level towards attaining a reproductively healthy society.

Counselling and creating awareness among people about reproductive organs, adolescence and associated changes, safe and hygienic sexual practices, sexually transmitted diseases (STDs) including AIDS, etc., is the primary step towards reproductive health. Providing medical facilities and care to the problems like menstrual irregularities, pregnancy related aspects, delivery, medical termination of pregnancy, STDs, birth control, infertility, post natal child and maternal management is another important aspect of the Reproductive and Child Health Care programmes.

An overall improvement in reproductive health has taken place in our country as indicated by reduced maternal and infant mortality rates, early detection and cure of STDs, assistance to infertile couples, etc. Improved health facilities and better living conditions promoted an explosive growth of population. Such a growth necessitated intense propagation of contraceptive methods. Various contraceptive options are available now such as natural, traditional, barrier, IUDs, pills, injectables, implants and surgical methods. Though contraceptives are not regular requirements for reproductive health, one is forced to use them to avoid pregnancy or to delay or space pregnancy.

Medical termination of pregnancy is legalised in our country. MTP is generally performed to get rid of unwanted pregnancy due to rapes, casual relationship, etc., as also in cases when the continuation of pregnancy could be harmful or even fatal to either the mother, or the foetus or both.

Diseases or infections transmitted through sexual intercourse are called Sexually Transmitted Diseases (STDs). Pelvic Inflammatory Diseases (PIDs), still birth, infertility are some of the complications of them. Early detection facilitate better cure of these diseases. Avoiding sexual intercourse with unknown/multiple partners, use of condoms during coitus are some of the simple precautions to avoid contracting STDs.

Inability to conceive or produce children even after 2 years of unprotected sexual cohabitation is called infertility. Various methods are now available to help such couples. *In Vitro* fertilisation followed by transfer of embryo into the female genital tract is one such method and is commonly known as the 'Test Tube Baby' Programme.